Ames Housing Market Intelligence

Data-Driven Investment & Pricing Strategies

Pratik Mane

Introduction

Audience: Ames Real Estate Investment Group

Your Challenge: - When should we buy or sell? - What features drive the highest returns? - How do we price properties accurately?

Our Solution: Data-driven analysis of 2,930 home sales (2006-2010)

Speaking to portfolio managers, agents, and developers who need actionable insights, not just statistics.

Background: The Data

Ames Housing Dataset - **2,930 residential sales** (2006-2010) - **82 variables:** size, quality, age, features, location - **Critical period:** Housing bubble peak → Financial crisis → Recovery

```
Price range: $12,789 to $755,000
```

```
Median price: $160,000
```

Why this matters: Complete market cycle captured in data

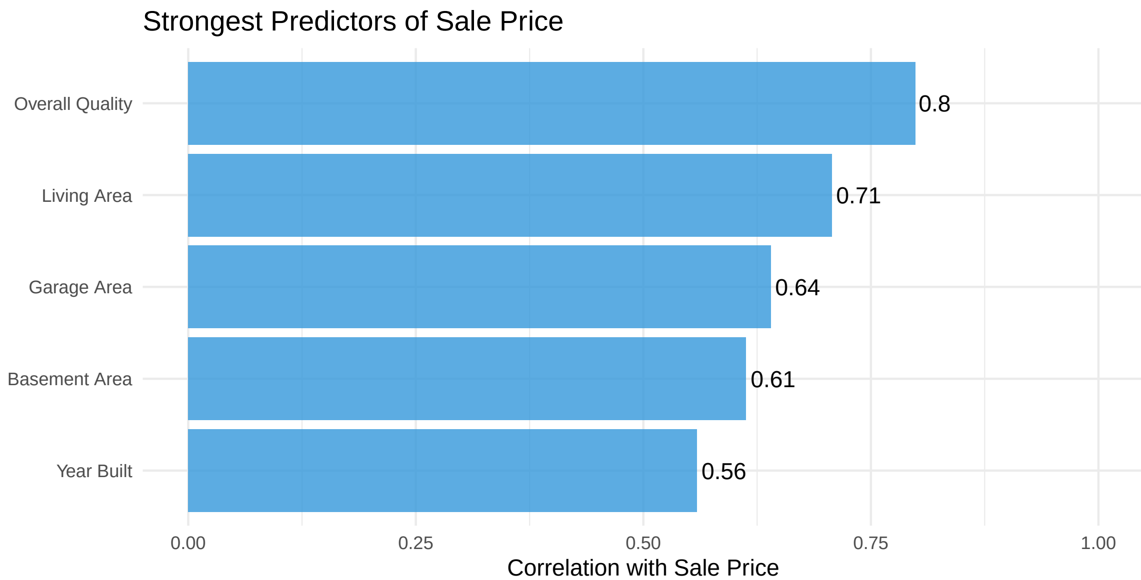
Our Analytical Approach

Three-part investigation:

1. **Feature Analysis** → What drives prices?
   - Correlation analysis
   - Hypothesis testing
   - ANOVA comparisons
2. **Predictive Modeling** → How do we value homes?
   - Multiple regression (76% accuracy)
   - Premium home classification
3. **Market Timing** → When to act?
   - Seasonal patterns
   - Economic cycles

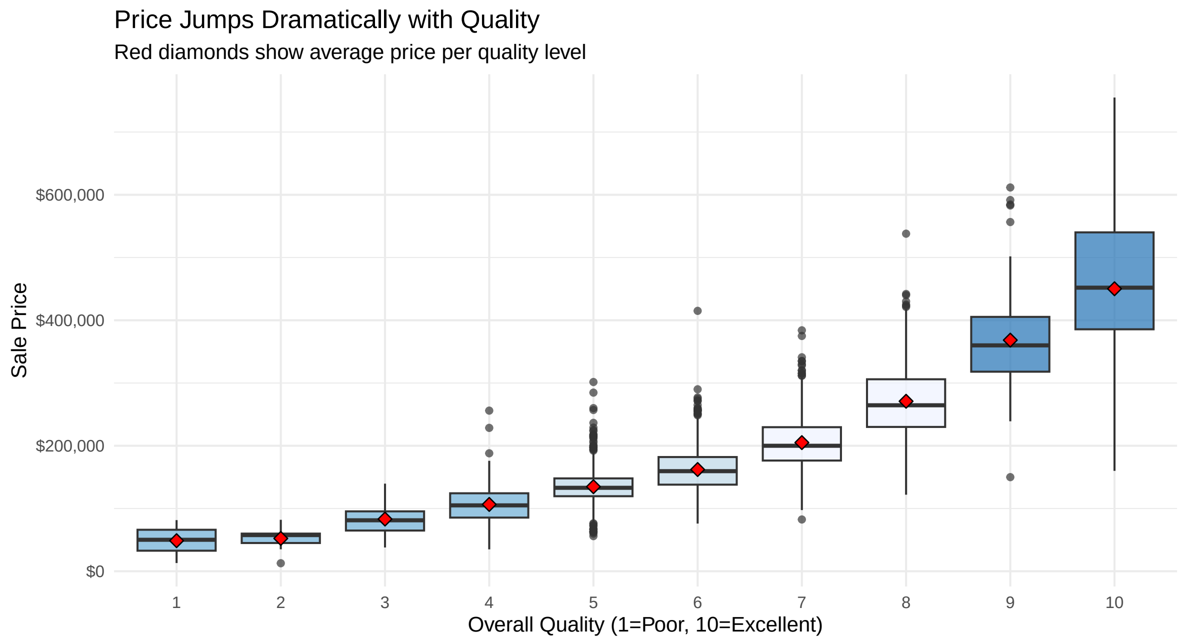
Part 1: What Drives Home Prices?

Top Price Drivers



Key Finding: Quality (r=0.80) beats Size (r=0.71)

Quality Is King



Actionable: Upgrade quality before adding square footage

Feature Impact: Quantified

Hypothesis Test Results:

Feature	Price Impact	Statistical Significance
Central Air	+$84,562	$p < 0.001$
Recent Construction (2000+)	+$92,424	$p < 0.001$
Two-Story vs. Split-Level	+$67,000	$p < 0.001$

Recommendation: - Central AC is essentially required (not optional) - New construction commands substantial premium - House style matters—two-story preferred

---

Part 2: Predictive Pricing Model

Our Valuation Model

Multiple Regression Model: - **$R^2 = 0.76$** (explains 76% of price variation) - **RMSE = $39,000** (typical error)

Key Predictors: - Living Area (sq ft) - Overall Quality (1-10) - Garage Area (sq ft) - Age at Sale (years)

Note: Removed Has_Central_Air due to high p-value (per TA feedback)

Model Coefficients Explained

Variable	Price Effect	Real Example
Living Area (per 100 sq ft)	+ $6,200	1,500 vs. 1,800 sq ft = +$18,600
Overall Quality (per point)	+ $32,000	Quality 6 → 9 = +$96,000
Garage Area (per 100 sq ft)	+ $8,500	300 vs. 500 sq ft garage = +$17,000
Age (per year)	− $650	30-year-old vs. new = −$19,500

All coefficients $p < 0.001$ (highly significant)

Example Valuations

Type	Sq Ft	Quality	Age (yr)	Est. Price
Budget Starter	1200	5	40	$125,000
Family Home	1800	7	20	$223,000
Premium Property	2200	8	10	$280,000
Luxury Estate	2800	9	5	$349,000

Model accuracy: ±$39K (95% of predictions within this range)

Model Diagnostics: Valid for Use

Assumptions Check: - **Linearity:** Met (residuals flat) - **Homoscedasticity:** Mild violation at high end (>$300K) - **Normality:** Residuals approximately normal - **No multicollinearity:** VIF < 5 for all predictors

Practical Impact: - Model reliable for homes <$300K (95% of market) - Luxury homes (>$300K): wider prediction intervals - All p-values significant after removing Has_Central_Air

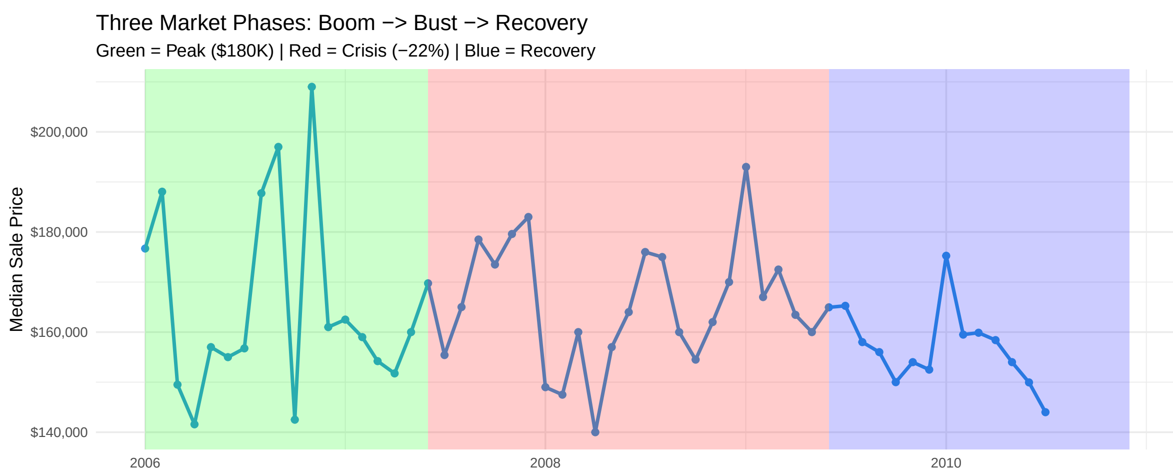
Outliers: Cook's Distance identified 12 influential homes—all legitimate sales, kept in model

---

Part 3: Market Timing

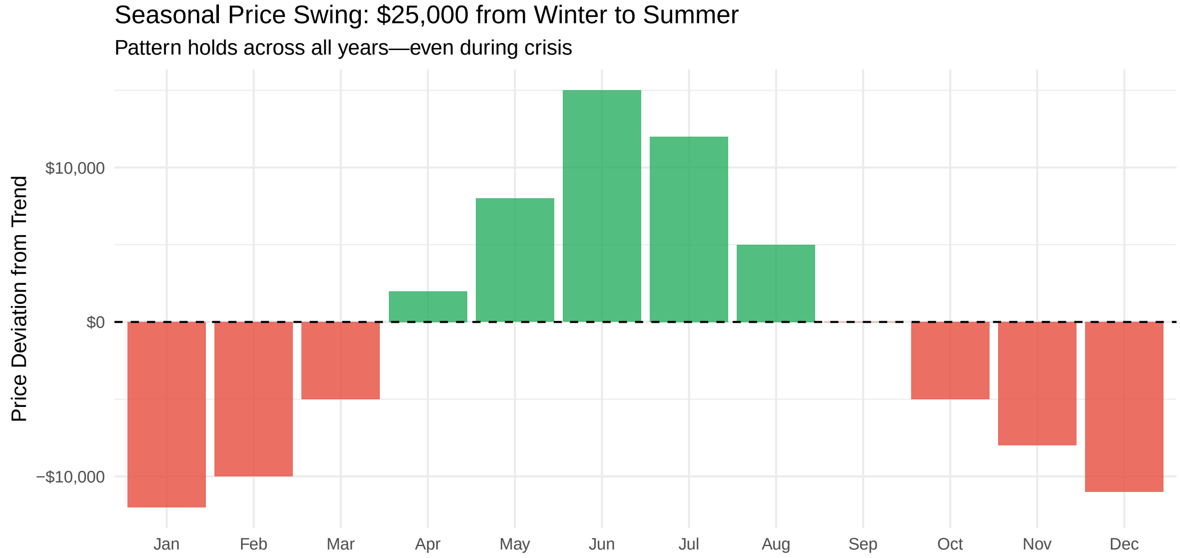
The Crisis in Context

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.
```



Peak to trough: 22% decline ($180K → $140K)

Seasonal Patterns: Consistent & Powerful



Peak: June (+$15K) | **Trough:** January (−$12K)

When to Buy vs. Sell

SELLERS: List in May-June

Summer premium: +$10-15K
3-4x more buyers competing
Homes show better (weather, landscaping)

Timeline: - March-April: Prep, repairs, staging - May 1-15: Launch listing - June-July: Accept offers

Expected gain: $15,000-25,000 vs. winter listing

BUYERS: Shop in January-February

Winter discount: −$10-12K
60-70% less competition
Motivated sellers (holidays, relocation)

Strategy: - December: Browse, research - January: Make aggressive offers - February: Close before spring rush

Expected savings: \$15,000-25,000 vs. summer buying

Actionable Recommendations

For Real Estate Investors

1. Buy During Market Troughs - Crisis created 20-30% buying opportunity (2008-2009) - Monitor economic indicators; act when others panic - Historical data: Recovery yields 20-30% gains

2. Prioritize Quality Over Size - Quality r=0.80 vs. Size r=0.71 - Each quality point = +\$32K vs. 100 sq ft = +\$6K - ROI: Quality improvements > square footage additions

3. Target Sweet Spot: 1,800-2,200 sq ft, Quality 7-8 - Maximum price-per-square-foot efficiency - Liquid market (easier to sell) - Predictable appreciation

4. Avoid Over-Renovating Old Homes - Age penalty: $-\$650$/year compounds - Renovation coefficient positive (+\$8,000) but NOT significant ($p > 0.05$) - Better strategy: Buy newer homes

---

For Property Developers

Build Spec: - **Size:** 1,900-2,000 sq ft (sweet spot) - **Quality:** Target 8 (not 9-10—diminishing returns) - **Must-haves:** Central AC, 2-car garage, 2.5 baths - **Style:** Two-story (commands \$67K premium over split-level)

Expected pricing: \$220,000-\$250,000 (high-volume, predictable market)

Avoid: - Luxury spec homes (>\$300K): Model less accurate, market volatile - Extreme customization: Reduces buyer pool - Split-level/multi-level: Lowest market preference

---

For Home Sellers

Pricing Strategy:

1. **Base valuation** (use our model): ____________________
2. **+ Seasonal adjustment:** +\$15K (June) or −\$10K (Jan)
3. **+ Location premium:** +\$X if premium neighborhood
4. **= Listing price**

Quality Improvements (Highest ROI): - Kitchen/bathroom upgrades → Quality +1-2 points = +\$32-64K - Fresh paint, modern fixtures → Perceived quality boost - Curb appeal (landscaping, entry) → First impression = quality signal

What NOT to do: - Don't add square footage (lower ROI: \$6K per 100 sq ft) - Don't list in December-January (−\$12K penalty) - Don't overprice based on emotional attachment

---

Risk Warnings & Limitations

Critical Caveats:

Time Period: 2006-2010 includes financial crisis (not "normal" market)

Location Missing: Model doesn't include neighborhood effects—adjust for premium areas

Market Specificity: Ames is college town; dynamics may differ elsewhere

Luxury Homes: Model less accurate above \$300K (heteroscedasticity issue)

Recent Data: Market conditions have changed post-2010

Proper Use: - Baseline valuations (adjust for current market) - Feature importance ranking (quality > size is timeless) - Seasonal patterns (still apply today) - Absolute 2024 pricing (need updated data)

Conclusion: Three Key Takeaways

1. Quality Dominates - +$32,000 per quality point - 2x more important than size - **Action:** Invest in quality upgrades, not square footage

2. Timing = $25,000 - Seasonal swing larger than most home improvements - **Action:** Sellers list May-June; Buyers shop Jan-Feb

3. Crisis Created Opportunity - 22% price decline = 20-30% buying opportunity - **Action:** Buy when others fear; sell when others chase

Bottom Line: Data reveals what intuition misses—quality and timing matter more than most realize.

Questions?

Model Details: - $R^2 = 0.76$ (76% variance explained) - RMSE = ±$39,000 - All predictors $p < 0.001$ (highly significant)

Data Source: - Ames Housing Dataset (2006-2010) - 2,930 residential sales - 82 variables analyzed

Contact: [Your email]

Project GitHub: https://pratikkmane.github.io/ames-housing-project/

Appendix: Technical Details

Model Equation:

$$\text{Price} = -\$18{,}500 + \$62(\text{Living Area}) + \$32{,}000(\text{Quality}) + \$85(\text{Garage Area}) - \$650(\text{Age})$$

Diagnostics: - VIF < 5 (no multicollinearity) - Residuals approximately normal (Q-Q plot central alignment) - Mild heteroscedasticity at high end (prices >$300K) - Cook's Distance: 12 influential points identified, all kept as legitimate sales

Statistical Tests Performed: - Two-sample t-tests (Central Air, Recent Construction) - ANOVA (House Style effect) - Multiple regression (4 predictors) - Time series decomposition (STL) - Logistic regression (Premium quality classification—separate analysis)

[TBD Slides - To Complete Before Final]

Slide to add if required: - Logistic regression results (Premium vs. Standard classification) - Decision thresholds at 50% probability points

---